CS4.406 Term Project — Mid-Submission Report

How fast does a news recommender go stale?

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1 What we are studying

A news recommender learns from past clicks and articles, then stays frozen. But news articles are replaced within hours. We want to know how fast a frozen recommender gets worse, and which kind of signal gets worse first: **word matching** (BM25), **meaning matching** (a frozen multilingual MiniLM model), or **click history** (which articles were read by the same people). Recency and popularity are simple baselines.

How we measure it. Take a real click. List the articles published in the 48 hours before it. Rank the clicked article among them (metrics: MRR, and the **top-10 hit rate**, which is recall@10: the share of clicks where the clicked article lands in the top 10; both with bootstrap confidence intervals). Do this for models frozen on different days and compare each one with a model rebuilt the day before. The gap is the *decay*.

2 Data

	EB-NeRD (main)	**NPR** (second)	**Adressa** (optional)
Publisher, language	Ekstra Bladet, Danish	Globo, Portuguese	Norwegian news site
Days of clicks	22 (May 2023)	118 (Jan–Apr 2023)	90 (Jan–Mar 2017)
Clicks	113.4 M	17.6 M	35.2 M
Users	791,582	1,162,802	3,614,911
Typical age of a clicked article	1.4 h	13.8 h	5.8 h

All three were cleaned into small per-day click files inside the project, and the article text was embedded on the GPU. For EB-NeRD we use the click history of the validation users, because it is the one part whose volume and type of click stay the same across dates.

3 Pipeline

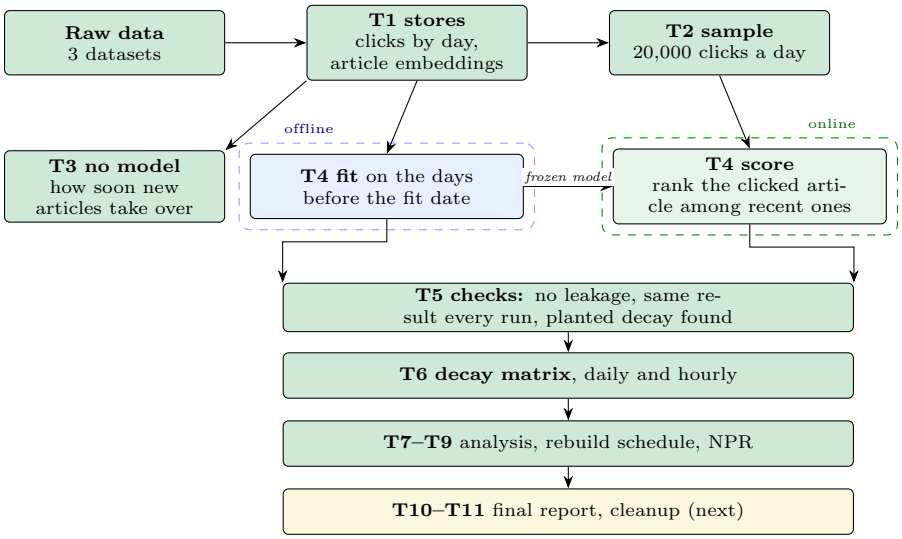


Figure 1: The pipeline. Green is done, yellow is next. A monitor watches time, memory and disk in every step.

4 What we found

Stages T1 to T9 are built, and all automated tests pass (including a planted decay that the checks recover). Results on EB-NeRD (14 fit days, about 17,000 clicks a day, 95% paired bootstrap intervals); "lag" is the days since the model was frozen. The user's profile is either *live* (last 50 clicks before the click) or *frozen* (clicks up to the fit date). The hit rates use the live profile:

	Top-10 hit rate, lag 1	Top-10 hit rate, lag 7	MRR lost by lag 7: frozen / live profile
Recency (newest first)	0.515 [0.513, 0.517]	0.528 [0.525, 0.530]	0 (by construction)
Word matching	0.061 [0.060, 0.062]	0.064 [0.063, 0.065]	0.0030 [0.0024, 0.0036] / 0.0001
Meaning matching	0.047 [0.046, 0.048]	0.047 [0.046, 0.048]	0.0007 [0.0003, 0.0011] / 0 (exactly)
Random shuffle	0.027 [0.026, 0.027]	0.027 [0.027, 0.028]	0
Click history, popularity	0.008	0.027	*coverage, see below*

- **Word and meaning matching go stale only through the user's profile.** With a live profile the loss is about 0; with a frozen profile word matching loses 0.0030 [0.0024, 0.0036] MRR by lag 7 (about 10%) and 0.0059 [0.0047, 0.0071] by lag 13 (about 24%). They are also weak: 2 to 3 times the shuffle. Recency is far ahead on this news site.
- **Click history and popularity fail by coverage, not by accuracy.** At lag 1 they can score only 12% of the clicked articles, and none at lag 7, because new articles have no clicks yet. Their scores fall to the shuffle level for that reason.
- **Hourly:** click history can score 79% of clicks one hour after the fit, 19% after 6 hours and 0.9% after 24 hours. Word matching loses 0.0023 [0.0020, 0.0026] MRR at 24 hours (about 7% of its score).
- **Rebuild interval:** with a live profile, rebuilding word matching daily instead of once gives 0.0325 against 0.0324 MRR, so a rebuild buys almost nothing; with a frozen profile, daily gives 0.0310 and once 0.0274; a click-history rebuild costs 22 s and about 7 GB.
- **No-model check:** at 6 hours, the share of new clicks that go to unseen articles is **90.8%** [88.5, 92.5] for EB-NeRD, **48.5%** [45.4, 51.5] for Adressa and **15.5%** [11.3, 20.1] for NPR.
- **NPR (second dataset)** ranks the signals differently: popularity hit rate 0.550 [0.547, 0.553], click history 0.249 [0.246, 0.251], recency 0.005 [0.004, 0.005]. Popularity gets *better* with an older fit, which we have not yet explained.
- **Check without never-clicked articles** (27% of the articles; this uses future clicks, so it is only a ceiling): every score rises (recency hit rate 0.515 to 0.634 [0.632, 0.636]), but the staleness loss does not change (word matching, live profile, 0.00013 to 0.00010 [0.00003, 0.00016] MRR at lag 7).

5 What went wrong, and what we changed

- Our first full run counted articles the user had already read. Our own check found that 15% of clicks were re-reads, which inflated word matching (MRR 0.057 instead of 0.014 without them). We deleted all results, dropped re-reads and the user's read articles from every pool, and reran everything.
- The click-history model needed 7.1 GB, not the 1.5 GB we planned, so only 2 can run at a time.
- Two hourly units were marked complete but had missing files. We found it when the analysis failed, and recomputed them.

6 Remaining work

Final report and AI log (T10), cleanup and a frozen copy of the results (T11). Not done: a hybrid model that falls back to recency when the others cannot score, Adressa, and the cause of popularity improving with an older fit on NPR.